The documentation and process conversion measures necessary to comply with this revision shall be completed by 10 January 1998.

INCH POUND

MIL-PRF-19500/548C
10 October 1997
SUPERSEDING
MIL-S-19500/548B
2 August 1993

PERFORMANCE SPECIFICATION SHEET

COUPLER, OPTO ELECTRONIC, SEMICONDUCTOR DEVICE, SOLID STATE
TYPES 4N47, 4N47A, 4N48, 4N48A, 4N49, 4N49A, 4N47U, 4N48U, AND 4N49U
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for solid state optically coupled isolators in which a gallium aluminum arsenide diode light source is optically coupled to a silicon NPN phototransistor. Four levels of product assurance are provided for each type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (4N47, 4N47A, 4N48, 4N48A, 4N49, and 4N49A)(similar to T0-99); see figure 2 (4N47U, 4N48U, and 4N49U).

1.3 Infrared-emitting diode maximum rating.

V_R	I_F 1/ 2/	I_P 3/
V dc	mA dc	A (pk)
2	40	1

1/ Derate linearly to 125° C at 0.67 mA/° C above +65° C.
2/ Minimum recommended operating I_F is 1 mA dc at +25° C.
3/ 1.0 µs pulse width, 300 pps.

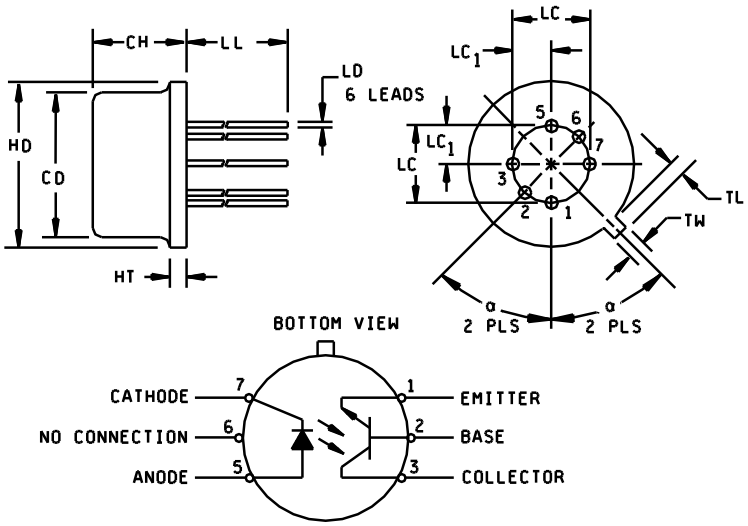
1.3.2 Phototransistor maximum rating.

V_{CEO}	V_{CBO}	V_{EBO}	P_T 1/	I_C
V dc	V dc	V dc	mW	mA dc
40	45	7	300	50

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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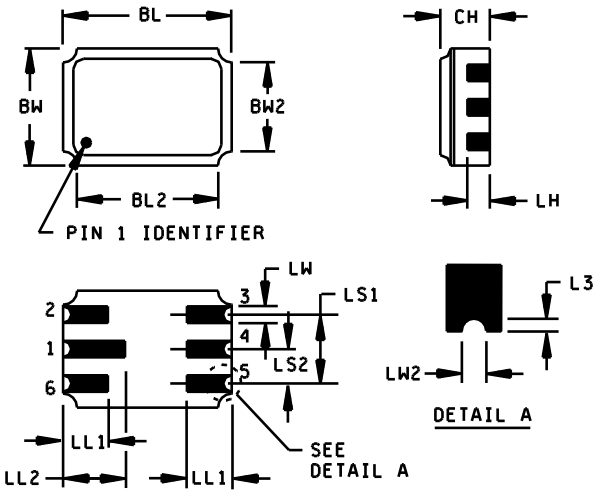


LTR	Dimensions				NOTES
	Inches		Millimeters		
	MIN	MAX	MIN	MAX	
CD	.305	.335	7.75	8.51	
CH	.155	.185	3.94	4.70	
HD	.335	.370	8.51	9.40	
HT	----	.040	----	1.02	
LC	.200 T.P.		5.08 T.P.		1
LC1	.100 T.P.		2.54 T.P.		1
LD	.016	.019	0.41	0.48	
LL	.500	.600	12.70	15.24	
TL	.029	.045	0.74	1.14	
TW	.028	.034	0.71	0.86	
α	45° T.P.		45° T.P.		

NOTES:
1. T.P. designates true position. Leads having maximum diameter .019 (0.48 mm) measured in gauging plane .054 +.001, -.000 (1.37 +0.03, -0.00 mm) below the seating plane of the device shall be within .007 (0.18 mm) of their true position relative to a maximum width tab.
2. Dimensions are in inches.
3. Metric equivalents are given for general information only.
4. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.
5. Device types 4N47A, 4N48A, and 4N49A have the collector isolated from the case.

FIGURE 1. Dimensions and configuration (4N47, 4N47A, 4N48, 4N48A, 4N49, and 4N49A).

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LTR	Dimensions			
	Inches		Millimeters	
	MIN	MAX	MIN	MAX
BL	.240	.250	6.10	6.35
BL2	----	.250	----	6.35
BW	.165	.175	4.19	4.44
BW2	----	.175	----	4.44
CH	.066	.080	1.68	2.03
LH	.026	.034	0.66	0.88
LL1	.060	.070	1.65	1.78
LL2	.082	.098	2.08	2.49
LL3	.003	.007	0.08	0.18
LS1	.095	.105	2.41	2.67
LS2	.045	.055	1.14	1.39
LW	.022	.028	0.56	0.71
LW2	.006	.022	0.15	0.56
N	6			
ND	3			
NE	0			

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Dimensions and configuration (4N47U, 4N48U, and 4N49U).

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1.3.3 Total device ratings.

T_J	T_{STG}	V_{IO}
°C	°C	V dc
-65 to +125	-65 to +125	1,000 V (max)

1.4 Primary electrical characteristics at T_A = +25°C.

1.4.1 LED (input) characteristics at T_A = +25°C.

Limits	I_R V_R = 2 V dc	V_F I_F = 10 mA dc	I_F 1/ V_F = 1.5 V dc
	µA dc	V dc	mA dc
Minimum		0.8	10
Maximum	100	1.5	

1/ Under no circumstances, shall I_F exceed the maximum rating of the device during the test.

1.4.2 Phototransistor (output) characteristics.

Limits	I_{C} off V_{CE} = 20 V	I_{CB} off V_{CB} = 20 V	$V_{(BR)CEO}$ I_C = 1 mA	$V_{(BR)CBO}$ I_E = 100 µA	$V_{(BR)EBO}$ I_E = 100 µA	h_{FE} V_{CE} = 5 V dc, I_C = 10 mA dc, I_F = 0
	nA	nA	V dc	V dc	V dc	
Minimum			40	45	7	100
Maximum	100	10				

1.4.3 Coupled (transfer) characteristics.

Limits	R_{IO} at 1,000 V see 4.5.5	C_{IO} V = 0 f = 1 MHz see 4.5.5	Phototransistor mode switching (see figure 3)						Photodiode mode switching (see figure 3)	
			t_r			t_f			t_r	t_f
			4N47 4N47A 4N47U	4N48 4N48A 4N48U	4N49 4N49A 4N49U	4N47 4N47A 4N47U	4N48 4N48A 4N48U	4N49 4N49A 4N49U		
	ohms	pF	µs	µs	µs	µs	µs	µs	µs	µs
Minimum	10^{11}									
Maximum		5	20	20	25	20	20	25	3	3

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1.4.3 Coupled (transfer) characteristics - continued.

Limits	$V_{CE(SAT)}$			$I_{C(ON)}$			$I_{CB(ON)}$
	4N47 4N47A 4N47U	4N48 4N48A 4N48U	4N49 4N49A 4N49U	4N47 4N47A 4N47U	4N48 4N48A 4N48U	4N49 4N49A 4N49U	
	I_F = 2 mA I_C = .5 mA	I_F = 2 mA I_C = 1 mA	I_F = 2 mA I_C = 2 mA	V_{CE} = 5 V I_F = 1 mA	V_{CE} = 5 V I_F = 1 mA	V_{CE} = 5 V I_F = 1 mA	V_{CB} = 5 V I_F = 10 mA
	V dc	V dc	V dc	mA dc	mA dc	mA dc	µA dc
Minimum				0.5	1.0	2.0	30.0
Maximum	0.3	0.3	0.3		5.0	10.0	

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

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3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

I_P - - - - - - - - - - - - - Peak forward pulse current.
$I_{C(ON)}$ - - - - - - - - - - - On-state collector current.
$I_{C(OFF)}$ - - - - - - - - - - - Off-state collector current.
R_{IO} - - - - - - - - - - - - - Input to output resistance.
V_{IO} - - - - - - - - - - - - - Input to output voltage.

3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figures 1(similar to T0-99) and 2(surface mount) herein. With the approval of the qualifying activity, this device is authorized to contain orgainic material for JANS.

3.3.1 Lead formation and finish for 4N47, 4N47A, 4N48, 4N48A, 4N49, and 4N49A,. Lead finish shall be solderable in accordance with MIL-STD-750, MIL-PRF-19500, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.3.2 Lead formation and finish for 4N47U, 4N48U, and 4N49U. Terminal pad shall be tungsten co-fired to the ceramic package. Terminal pad finish shall be gold plated or solder dipped. Where a choice of lead finish is desired, it shall be specified in the acquisition document (6.2).

3.3.3 For Eutectic alloyed die of less than 1.5 X 10-5 square inches of surface area. Die Shear process control shall be in accordance with the requirements for small area die in MIL-STD-883, method 2019.

3.3.3 JAN, JANTX, AND JANTXV ONLY. Die shear process control on epoxy mounted die is not required.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2).

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in paragraphs 4.4.2 and 4.4.3.

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV level
3	See 4.3.1	See 4.3.1
9	$I_{C(OFF)1}$ and h_{FE}	$I_{C(OFF)1}$ 100 percent read and record
10	See 4.3.2	See 4.3.2
11	I_R, $I_{C(OFF)1}$, h_{FE}, and $I_{C(ON)1}$; $\Delta I_{C(OFF)1}$ = 100% of initial value or = 25 nA dc whichever is greater; Δh_{FE} ±20% of initial reading.	I_R, $I_{C(OFF)1}$, h_{FE}, and $I_{C(ON)1}$; $\Delta I_{C(OFF)1}$ = 100% of initial value or 25 nA dc whichever is greater.
12	See 4.3.3	See 4.3.3
13	Subgroups 2 and 3 of table I herein; $\Delta I_{C(OFF)1}$ = 100% of initial value or 25 nA dc whichever is greater. Δh_{FE} = ±20 % of initial reading; $\Delta I_{C(ON)1}$ = ±25% of initial reading; ΔI_R = 100% of initial value or 25 μA dc, whichever is greater.	Subgroup 2 of table I herein; $\Delta I_{C(OFF)1}$ = 100% of initial value or 25 nA dc whichever is greater; Δh_{FE} = ±20% of initial reading; $\Delta I_{C(ON)1}$ = ±25% of initial reading ΔI_R = 100% of initial value or 25 µA dc, whichever is greater.

4.3.1 Thermal shock (temperature cycling). All devices shall be subjected to thermal shock (temperature cycling) in accordance with MIL-STD-750, method 1051, test condition B, except $T_{(min)}$ = -55° C; 10 cycles, time at temperature extremes = 15 minutes minimum.

4.3.1.1 Monitored thermal shock (temperature cycling). One cycle of monitored temperature cycling shall be performed on 100 percent of the devices. This test shall be performed any time after the completion of the thermal shock test specified or it may be the last of the 10 thermal cycles. All junctions shall be monitored for electrical continuity. Any discontinuity shall be cause for rejection of the device(s) under test. If 10 percent or more of the number of devices subjected to monitored temperature cycling fail, the entire lot shall be rejected as "TX", "TXV", or "S" types.

4.3.2 High temperature reverse bias (HTRB). All devices shall be subjected to high temperature reverse bias in accordance with MIL-STD-750, method 1039, test condition A, T_A = +125° C, I_F = 0, V_{CB} = 36 V dc for 48 hours minimum.

4.3.3 Power burn-in conditions. Power burn-in conditions are as follows: See figure 4 for burn-in circuit. V_{CC} = 20 V dc, V_{CE} = 10 ±5 V dc, I_F = 40 mA dc, P_T = 275 ±25 mW at T_A = +25° C ±3° C. No heat sink or forced air cooling directly on the devices shall be permitted.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein. (Endpoint electrical measurements shall be in accordance with the applicable steps of table III herein.)

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (endpoints) and delta requirements shall be in accordance with the applicable steps of table III herein.

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4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1056	Test condition B.
B3	1071	Fine leak: Condition G or H; Gross leak: Condition C
B3	2037	Bond strength: test condition A. Sample used in decap internal visual design verification may be used in bond strength and die shear. Sample is in units.
B4	----	High temperature isolation voltage test (see 4.5.2) and as follows: V_{ISO} = 150 V dc, T_A = +125° C, t = 24 hours minimum. n = 22, c = 0.
B4	1037	V_{CE} = 10 V dc, I_F = 20 mA dc, P_T = 275 ± 25 mW at T_A = +25° C ±3° C (see figure 4), t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced air cooling directly on the device shall be permitted.
B5	1027	V_{CE} = 10 V dc, T_A = +100° C ±3° C for 96 hours, P_T = 275 ± 25 mW, I_F = 20 mA dc (see figure 4), or adjusted as required by the chosen T_A to give an average lot T_J = +200° C. Marking legibility requirements shall not apply.
B5	2037	(Al-Au die interconnects only), test condition A.
B5	1018	Internal water vapor test, n = 3, c = 0.
B6	----	Not applicable.

4.4.2.2 Group B inspection, table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B2	1056	Temperature cycling: test condition B (25 cycles). Total test time = 72 hours maximum.
B2	1071	Fine leak - condition G or H, gross leak - condition C.
B3	----	High temperature isolation voltage test conditions: V_{ISO} = 150 V dc (see 4.5.2), T_A = +125° C, t = 24 hours minimum. n = 22, c = 0.
B3	1027	I_F = 20 mA dc, P_T = 275 ± 25 mW at T_A = +25° C ±3° C (see figure 4). No heat sink or forced air cooling directly on the device shall be permitted.
B4	2037	Test condition A.
B5 and B6	----	Not applicable.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table III herein.

Subgroup	Method	Condition
C2	1056	Thermal shock (temperature cycling): test condition A.
C2	2036	Terminal strength, test condition E.
C2	1071	Fine leak - condition G or H; gross leak - condition C.
C3	2016	Nonoperating; 1,500 G, t = 0.5 ms, 5 blows in each orientation: X_1, Y_1, and Y_2.
C3	2056	50 G minimum. Nonoperating.
C3	2006	Nonoperating, 30,000 G, X_1, Y_1, and Y_2, orientations.
C6	1026	I_F = 20 mA dc, P_T = 275 ±25 mW at T_A = +25°C ±3°C (see figure 4). No heat sink or forced air cooling directly on the device shall be permitted.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with MIL-PRF-19500 and table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions of pulse measurements shall be as specified in MIL-STD-750.

4.5.2 High temperature isolation voltage lot verification. This test shall be performed by shorting both the anode and cathode terminals of the LED (Light Emitting Diode) and shorting together the collector, emitter, and base terminals of the transistor. The sample shall be split into two approximately equal groups and the specified isolation voltage (V_{ISO}) applied between the LED terminals (positive) and transistor terminals (negative) of one group. The specified isolation voltage shall be repeated with the LED terminals (negative) and the transistor terminal (positive) for the remaining group. All voltages shall be applied at the specified ambient temperature.

4.5.3 Input LED tests. These tests shall be performed with the output transistor terminals open.

4.5.4 Output transistor tests. These tests shall be performed with the input LED terminals open.

4.5.5 Isolation and coupling capacitance tests. These tests shall be performed between both input terminals 5 and 7 shorted together and the output terminals (1, 2, and 3, see figure 1) shorted together.

4.6 Internal visual (PRECAP) inspection. Internal visual inspection shall be performed in accordance with MIL-STD-750, methods 2072 and 2073, and as specified herein. Method 2072 shall be used for inspecting transistor die and completed assembly. Method 2073 shall be used for inspecting light emitting diode.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical	2071					
Subgroup 2 2/						
Reverse current leakage	4016	V_R = 2 V dc (see 4.5.3)	I_R		100	µA dc
Forward voltage	4011	I_F = 10 mA dc (see 4.5.3)	V_{F1}	0.8	1.5	V dc
Forward current		V_F = 1.5 V dc (see 4.5.3)	I_F	10.0		mAdc
(Transistor characteristics)						
Collector to emitter breakdown voltage	3011	I_C = 1 mA dc, I_B = 0, I_F = 0 (see 4.5.4)	$V_{(BR)CEO}$	40		V dc
Collector to base breakdown voltage	3001	I_C = 100 µA dc, I_F = 0, I_E = 0 (see 4.5.4)	$V_{(BR)CBO}$	45		V dc
Emitter to base breakdown voltage	3026	I_E = 100 µA dc, I_C = 0, I_F = 0 (see 4.5.4)	$V_{(BR)EBO}$	7		V dc
Off-state collector current (phototransistor mode)	3041	V_{CE} = 20 V dc, I_B = 0, I_F = 0, condition D (see 4.5.4)	$I_{C(OFF)1}$		100	nA dc
Off-state collector current (phototransistor mode)		V_{CB} = 20 V dc, I_F = 0, (see 4.5.4)	$I_{C(OFF)}$		10	nA dc
DC current gain	3076	V_{CE} = 5 V; I_C = 10 mA dc, I_F = 0 (see 4.5.4)	h_{FE}	100		
(Coupler characteristics)						
On-state collector current (phototransistor mode)		V_{CE} = 5 V dc, I_F = 1 mA dc	$I_{C(ON)1}$			
4N47, 4N47A, 4N47U				0.5		mA dc
4N48, 4N48A, 4N48U				1.0	5.0	mA dc
4N49, 4N49A, 4N49U				2.0	10.0	mA dc
On-state collector current (phototransistor mode)		V_{CB} = 5 V dc, I_F = 10 mA dc	$I_{CB(ON)}$	30.0		µAdc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued 2/						
Collector to emitter saturation voltage			$V_{CE(SAT)}$			
4N47, 4N47A, 4N47U		I_C = 0.5 mA dc, I_F = 2 mA dc			0.3	V dc
4N48, 4N48A, 4N48U		I_C = 1 mA dc, I_F = 2 mA dc			0.3	V dc
4N49, 4N49A, 4N49U		I_C = 2 mA dc, I_F = 2 mA dc			0.3	V dc
Input to output internal resistance		$\lvert V_{IN-OUT} \rvert$ = 1 k V dc (see 4.5.4)	R_{IO}	10^{11}		ohms
Subgroup 3 2/						
High temperature operation		T_A = +100°C				
(Transistor characteristics)						
Off-state collector current (phototransistor mode)	3041	V_{CE} = 20 V dc, I_F = 0, I_B = 0, condition D, (see 4.5.4)	$I_{C(OFF)2}$		100	µAdc
(Coupler characteristics)						
On-state collector current (phototransistor mode)	3041	V_{CE} = 5 V dc, I_F = 2 mA dc	$I_{C(ON)2}$			
4N47, 4N47A, 4N47U				0.5		mA dc
4N48, 4N48A, 4N48U				1.0		mA dc
4N49, 4N49A, 4N49U				2.0		mA dc
(LED characteristics)						
Forward voltage	4011	I_F = 10 mA dc,(see 4.5.3)	V_{F2}	0.7	1.3	V dc
Low temperature operation		T_A = -55°C				
(Coupler characteristics)						
On-state collector current (phototransistor mode)	3041	V_{CE} = 5 V dc, I_F = 2 mA dc	$I_{C(ON)3}$			
4N47, 4N47A, 4N47U				0.7		mA dc
4N48, 4N48A, 4N48U				1.4		mA dc
4N49, 4N49A, 4N49U				2.8		mA dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued 2/						
(LED characteristics)						
Forward voltage	4011	I_F = 10 mA dc,(see 4.5.3)	V_{F3}	1.0	1.7	V dc
Subgroup 4 2/						
Input to output capacitance		f = 1 MHz, (see 4.5.5), $\lvert V_{IN-OUT} \rvert$ = 0	C_{IO}		5	pF
Rise time (phototransistor mode)		V_{CC} = 10 V dc, I_F = 5 mA dc, R_L = 100 ohms (see figure 3)	t_r			
4N47, 4N47A, 4N47U					20	µs
4N48, 4N48A, 4N48U					20	µs
4N49, 4N49A, 4N49U					25	µs
Fall time (phototransistor mode)		V_{CC} = 10 V dc, I_F = 5 mA dc, R_L = 100 ohms (see figure 3)	t_f			
4N47, 4N47A, 4N47U					20	µs
4N48, 4N48A, 4N48U					20	µs
4N49, 4N49A, 4N49U					25	µs
Rise time (photodiode mode)		V_{CC} = 10 V dc, I_F = 5 mA dc, R_L = 100 ohms (see figure 3)	t_r		3	µs
Fall time (photodiode mode)		V_{CC} = 10 V dc, I_F = 5 mA dc, R_L = 100 ohms (see figure 3)	t_f		3	µs
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

2/ All devices required by the specified sampling plan are subjected to subgroups 2, 3, and 4 combined. (JANS only)

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TABLE II. Group E inspections (all quality levels) for qualification only.

Inspections	MIL-STD-750		Quality conformance inspection sample size
	Method	Conditions	
Subgroup 1		Test condition B, 500 cycles	45 devices, c = 0
Thermal shock (temperature cycle)	1051		
Hermetic seal	1071		
a. Fine leak		Condition G or H for fine leak	
b. Gross leak		Condition C for gross leak	
Electrical measurements		See table III, steps 1, 2, and 3	
Subgroup 2			45 devices, c = 0
High temperature reverse bias	1039	For all devices with organic material	
Electrical measurements		See table III, steps 1, 2, and 3	

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TABLE III. Groups A, B, C, and E electrical end-point measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Off-state collector current	3401	V_{CE} = 20 V dc; I_F = 0, condition D,(see 4.5.4)	$I_{C(OFF)1}$		100	nA dc
2.	On-state collector current		V_{CE} = 5 V dc, I_F = 1 mA dc	$I_{C(ON)1}$			
	4N47, 4N47A, 4N47U				0.5		mA dc
	4N48, 4N48A, 4N48U				1.0	5.0	mA dc
	4N49, 4N49A, 4N49U				2.0	10.0	mA dc
3.	Collector to emitter saturation voltage			$V_{CE(SAT)}$			
	4N47, 4N47A, 4N47U		I_C = 0.5 mA dc, I_F = 2 mA dc			0.3	V dc
	4N48, 4N48A, 4N48U		I_C = 1 mA dc, I_F = 2 mA dc			0.3	V dc
	4N49, 4N49A, 4N49U		I_C = 2 mA dc, I_F = 2 ma dc			0.3	V dc
4.	DC current gain	3076	V_{CE} = 5 V dc, I_C = 10 mA dc, I_F = 0 (see 4.5.4)	h_{FE}	100		
5.	Reverse current leakage	4016	V = 2 V dc (see 4.5.3)	I_R		100	µA dc
6.	Off-state collector current	3041	V_{CE} = 20 V dc, I_F = 0, condition D, (see 4.5.4)	$\Delta I_{C(OFF)1}$ 4/	±100% of initial value or 25 nA dc, whichever is greater.		
7.	On-state collector current		V_{CE} = 5 V dc, I_F = 1 mA dc	$\Delta I_{C(ON)1}$ 4/	±25% change from initial reading.		
8.	DC current gain	3076	V_{CE} = 5 V dc, I_C = 10 mA dc, I_F = 0 (see 4.5.4)	Δh_{FE} 4/	±25% change from initial reading.		
9.	Reverse current leakage	4016	V = 2 V dc, (see 4.5.3)	I_R 4/	±100% of initial value or 25 µA dc, whichever is greater.		
10.	Collector to emitter saturation voltage		As in step 3.	$\Delta V_{CE(SAT)}$ 4/	±50 mV dc change from previous measured value.		

See footnotes at end of table.

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TABLE III. Groups A, B, C, and E electrical end-point measurements - Continued. 1/ 2/ 3/

1/ The electrical measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:

a. Subgroup 3, see table III herein, steps 1, 2, 3, and 5.

b. Subgroup 4, see table III herein, steps 1, 2, 3, and 5.

c. Subgroup 5, see table III herein, steps 1, 2, 4, 5, 6, 7, 8, 9, and 10.

2/ The electrical measurements for table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 are as follows:

a. Subgroup 2, see table III herein, steps 1, 2, and 3.

b. Subgroup 3, see table III herein, steps 1, 2, 5, 6, and 7.

c. Subgroup 6, see table III herein, steps 6 and 7.

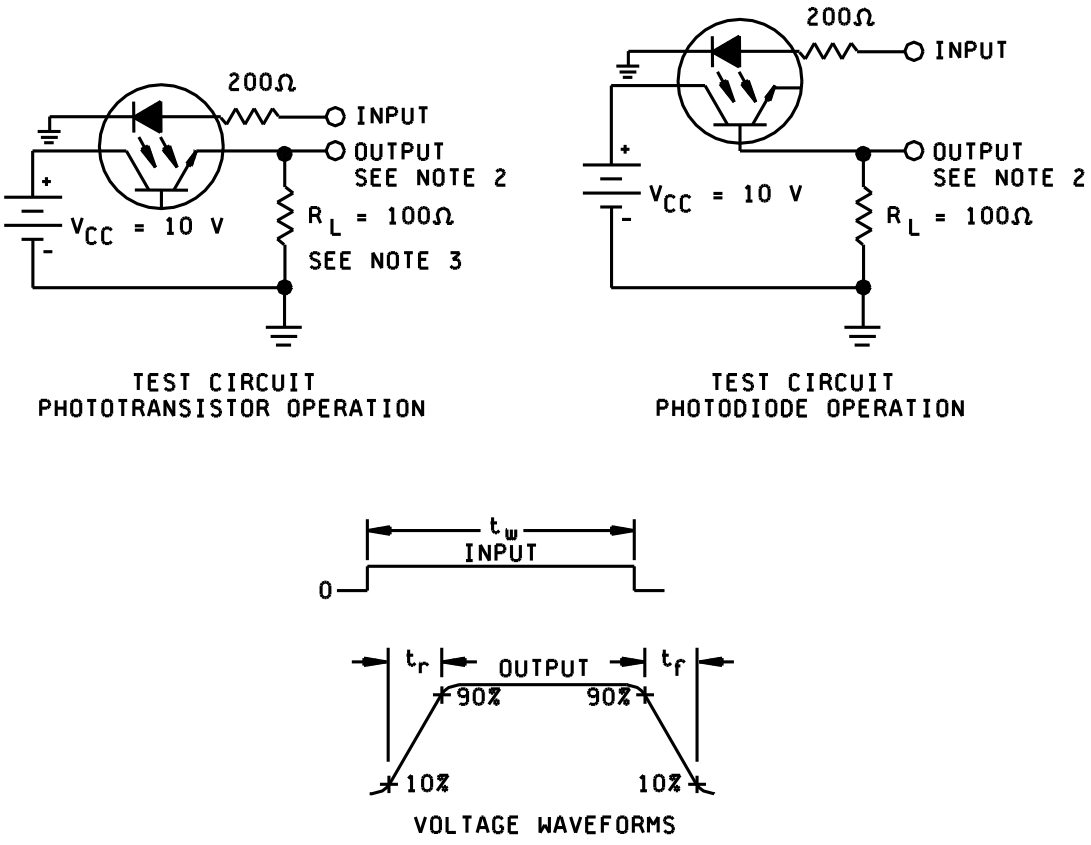
3/ The electrical measurements for table VII of MIL-PRF-19500 are as follows:

a. Subgroup 2, see table III herein, steps 1, 2, 3, and 5 (JANS) and steps 1 and 2 (JAN, JANTX, and JANTXV).

b. Subgroup 3, see table III herein, steps 2 and 3.

4/ Devices which exceed the group A limits for test shall not be acceptable for shipment.

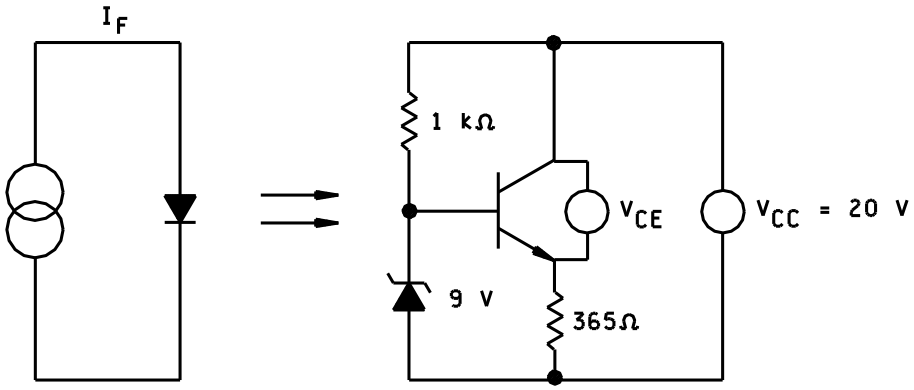
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NOTES:
1. The input waveform is supplied by a generator with the following characteristics: Z_{OUT} = 50 Ω, $t_r \leq$ 15 ns, duty cycle = 1%, t_w = 100 µs.
2. The output waveform is monitored on an oscilloscope with the following characteristics: $t_r \leq$ 12 ns, R_{in} = 1 MΩ, C_{in} = 20 pF.
3. Adjust amplitude of input for: I_F = 5 mA dc.

FIGURE 3. Switching times.

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NOTE: I_F = 20 mA dc for operation life tests.
I_F = 40 mA dc for burn-in of TX, TXV, and S.

FIGURE 4. Operation life test burn-in circuit.

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4.6.1 Gallium arsenide or gallium aluminum arsenide light emitting diode die inspection. The die shall be inspected under 100X magnification.

4.6.2 Silicon phototransistor visual inspection. The die shall be inspected under 100X magnification.

4.6.3 Die position. For applicable constructions, emitter die shall not over-hang the sensor by more than 33 percent of the width of the emitter die.

4.6.4 Die connection. All connections between die assemblies and header posts shall be intact (wires). The connections between the collector post and the header eyelet shall be intact.

4.6.5 Ball support. For applicable constructions, ball bond on cathode contact of light emitter die shall be fully supported underneath with glass coupling preformed and shall not extend to within more than one mil of the edge of the die.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agencys' automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Product assurance level and type designator.

c. Packaging requirements (see 5.1).

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VAT, Columbus, OH 43216.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, OS, SH
Air Force - 13, 19, 85, 99

Preparing activity:
DLA - CC

(Project 5980-0014)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/548C	**2. DOCUMENT DATE (YYMMDD)** 97/10/08

3. DOCUMENT TITLE COUPLER, OPTO ELECTRONIC, SEMICONDUCTOR DEVICE, SOLID STATE TYPES 4N47, 4N48, 4N49, 4N47U, 4N48U, AND 4N49U JAN, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 614-692-0510 (2) AUTOVON 850-0510
c. ADDRESS (Include Zip Code) Defense Supply Center Columbus ATTN: DSCC-VAT Columbus, Ohio 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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